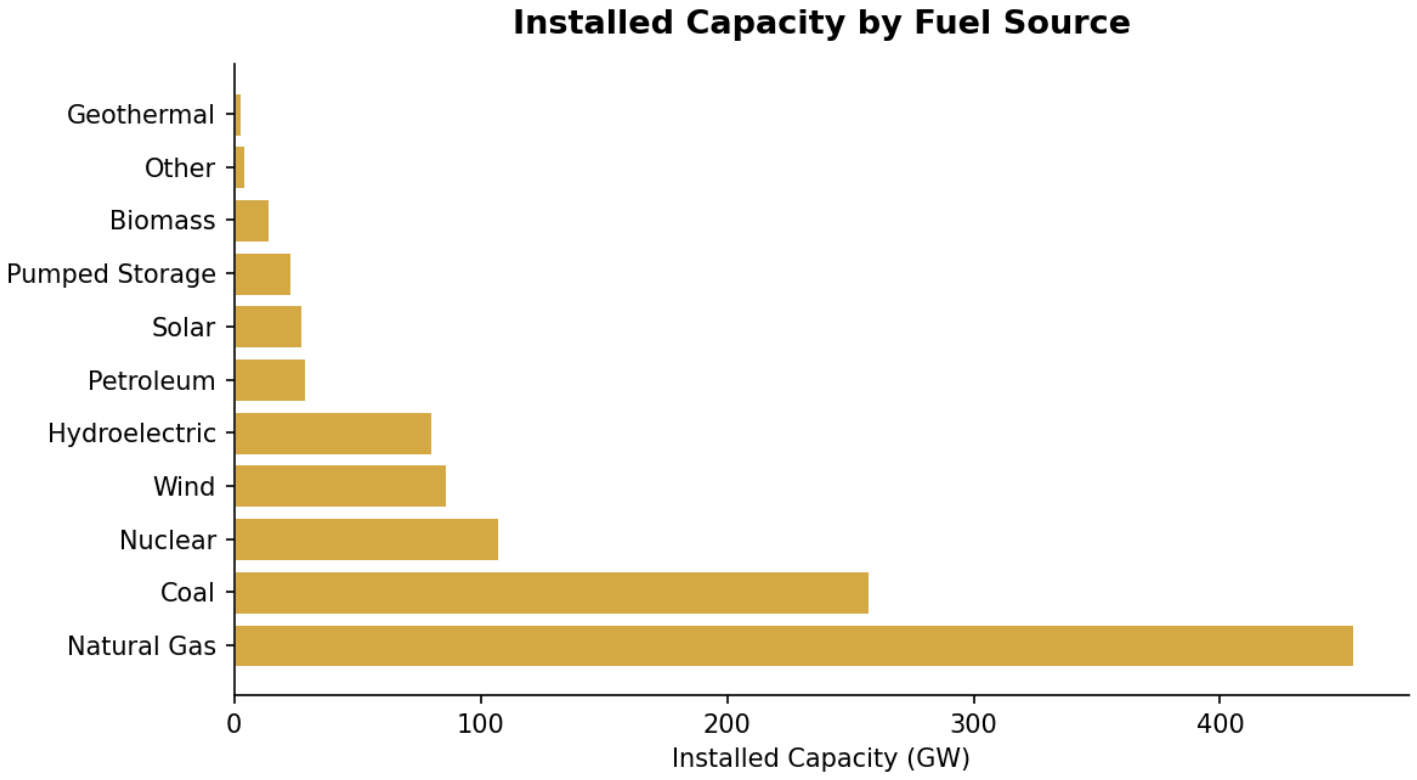
# America Has a Lot of Power Plants. Most of Them Barely Matter.

The United States has over 8,600 power plants, sprawling across every state and nearly every county in the nation. But a look at federal data reveals a stark imbalance: a small number of enormous facilities generate the bulk of the country's electricity, while thousands of tiny plants contribute almost nothing to the grid.

The data, drawn from the Energy Information Administration's survey of power plants, paints a portrait of an energy system caught between its fossil-fueled past and a renewable future that is arriving unevenly -- and in some places, hardly at all.

## The Fossil Fuel Backbone

Natural gas dominates American power generation, and it isn't close. Gas-fired plants account for about 54 percent of the nation's installed capacity, or roughly 554,000 megawatts. Coal, despite decades of decline, still represents nearly a quarter of total capacity across 337 plants producing about 258,000 megawatts.



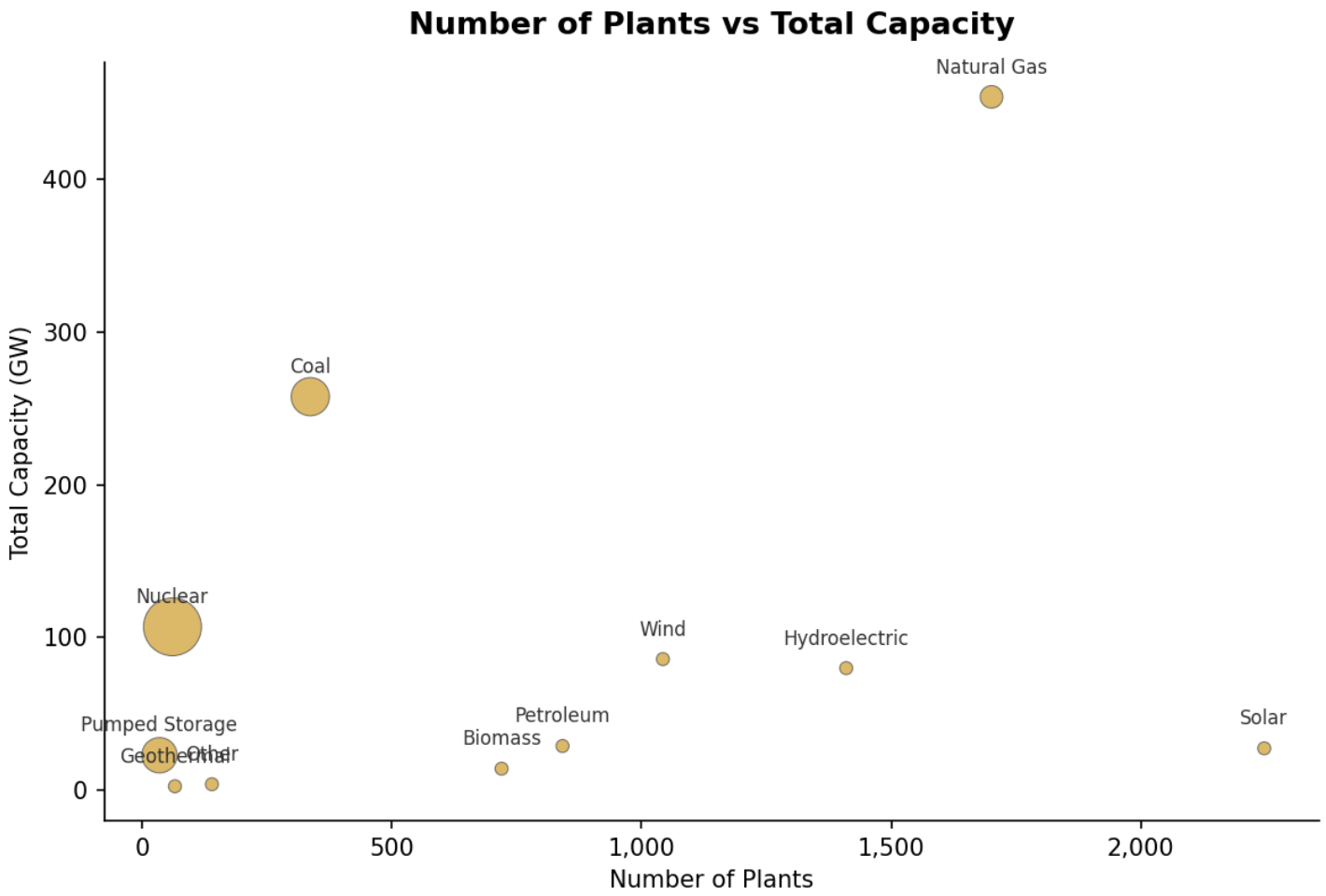

Together, fossil fuels -- natural gas, coal and petroleum -- make up about 68 percent of installed capacity nationwide. Nuclear energy, generated by just 45 plants, adds another 22 percent.

## Renewables: Many Plants, Modest Power

The renewable story is one of quantity over quality -- at least for now. Solar leads the country in sheer

number of facilities, with 2,247 plants, more than any other single source. Wind farms number over a thousand.

But those 2,247 solar installations produce just 27,400 megawatts combined -- roughly 2.5 percent of the national total. The average solar plant clocks in at about 12 megawatts. By comparison, the average nuclear plant produces over 2,100 megawatts, making a single reactor complex the equivalent of more than 172 solar farms.



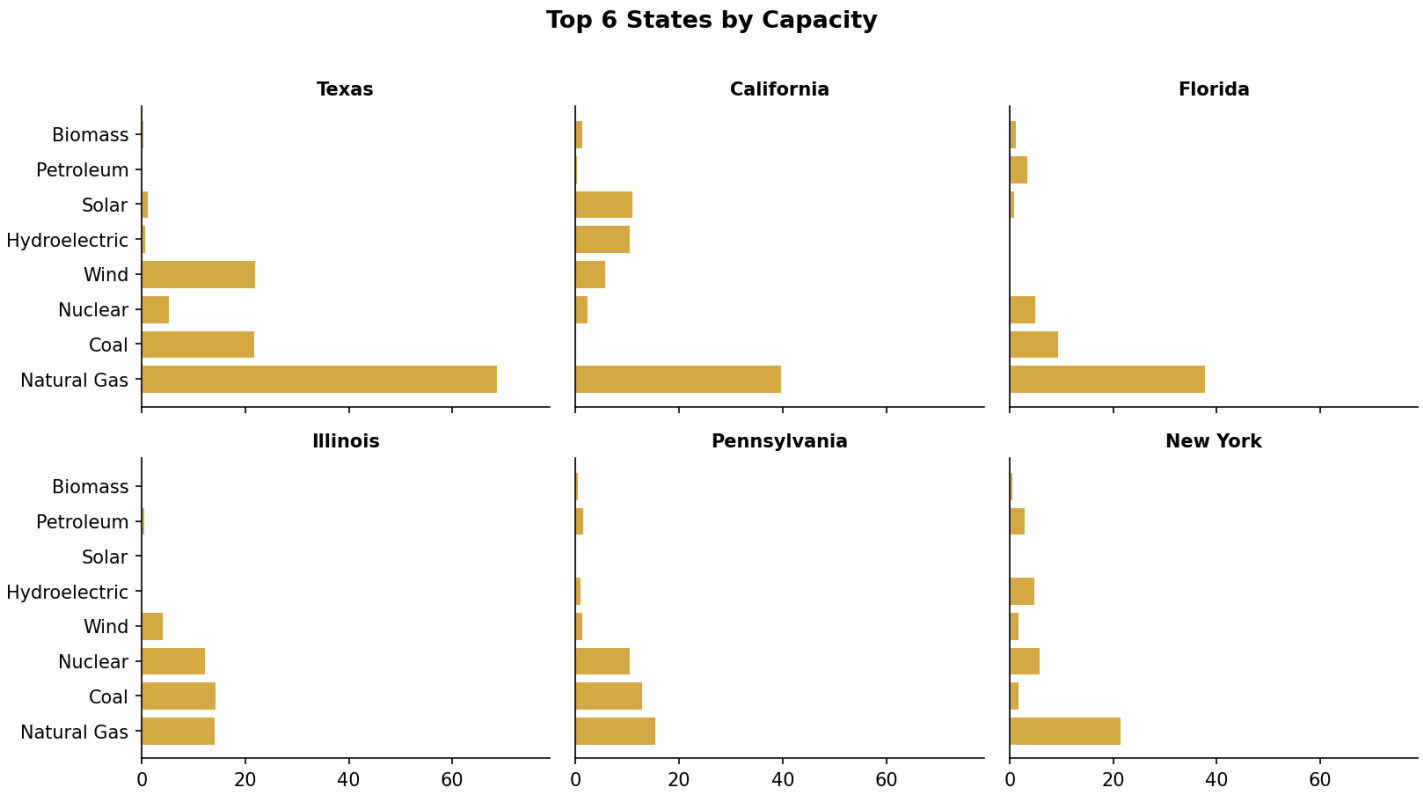

Taken together, all renewable sources -- solar, wind, hydro, geothermal and biomass -- account for about 4,388 facilities but nearly 30 percent of installed capacity. They outnumber fossil fuel plants nearly two to one, yet still trail far behind in actual power output.

## Texas: Big in Everything

No state has more installed generating capacity than Texas, which leads the nation at roughly 120,000 megawatts across 323 plants. Natural gas makes up 57 percent of Texas's capacity, followed by wind at 18 percent and coal at just 18 percent.

That wind figure is notable: Texas has more installed wind capacity than any other state, at about 21,900 megawatts. Kansas ranks second, with about 10,500 megawatts of wind capacity.

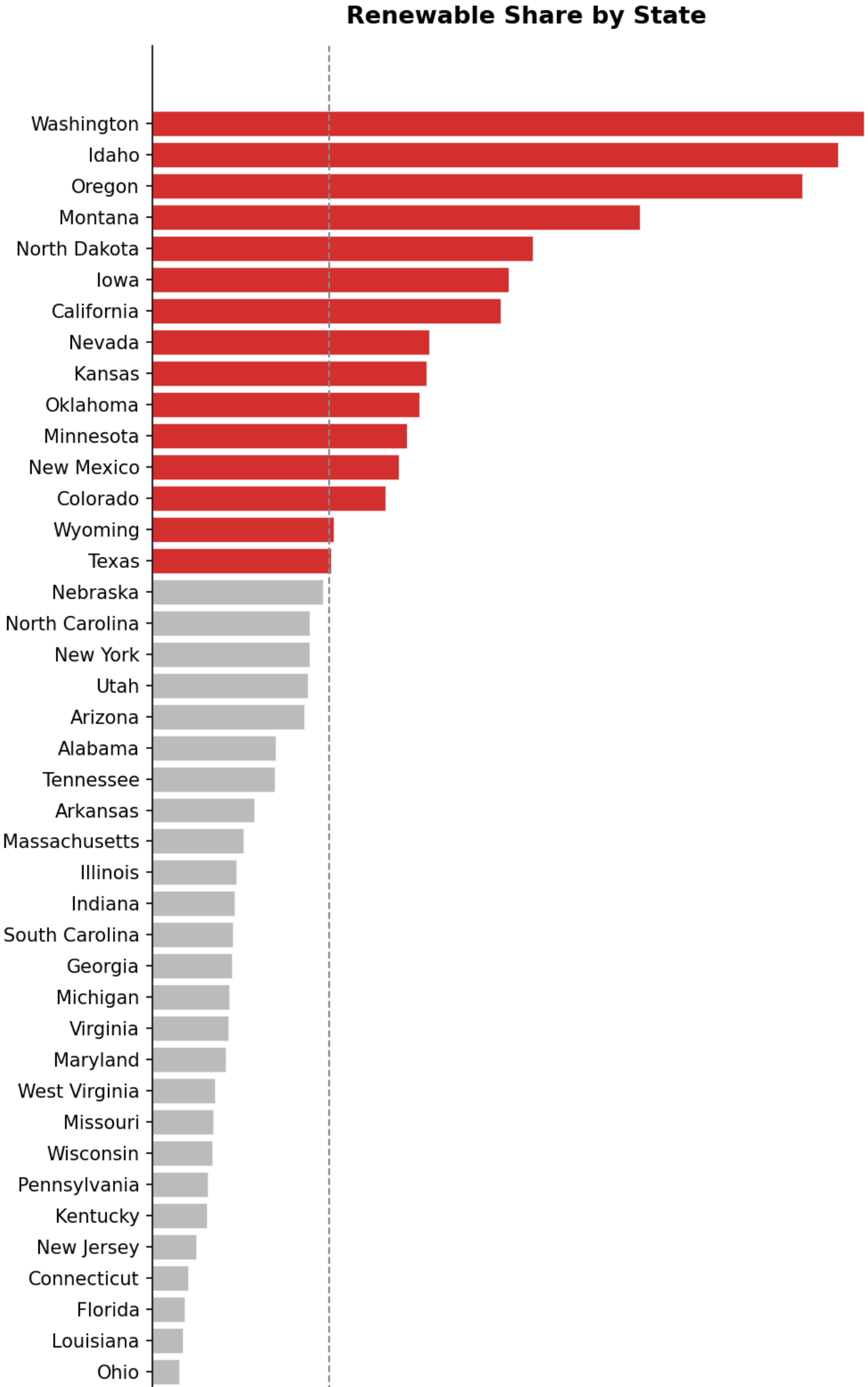
California has the most plants of any state -- 1,340 -- but ranks second in total capacity at about 77,000 megawatts. Roughly half of California's capacity comes from renewable sources, the highest share among the ten largest energy-producing states.




## The Geography of Inertia

The renewable gap is starkest in the Southeast and parts of the Midwest. Florida, with nearly 58,000 megawatts of installed capacity, gets just 3.7 percent from renewables. Mississippi sits at 2.7 percent. Ohio and Louisiana are both below 4.0 percent.

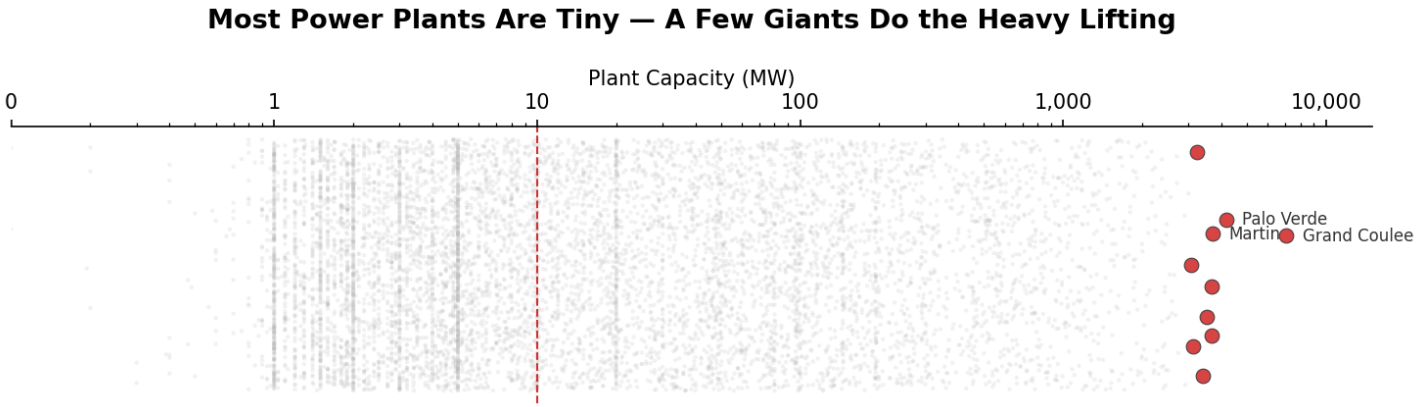
The massive Grand Coulee Dam -- the single largest power plant in the country at 7,079 megawatts -- sits in Oregon, which leads large states with 74 percent of its capacity from renewable sources. Washington follows close behind at 81 percent.


Renewable Share by State
Washington
Idaho
Oregon
Montana
North Dakota
Iowa
California
Nevada
Kansas
Oklahoma
Minnesota
New Mexico
Colorado
Wyoming
Texas
Nebraska
North Carolina
New York
Utah
Arizona
Alabama
Tennessee
Arkansas
Massachusetts
Illinois
Indiana
South Carolina
Georgia
Michigan
Virginia
Maryland
West Virginia
Missouri
Wisconsin
Pennsylvania
Kentucky
New Jersey
Connecticut
Florida
Louisiana
Ohio

## A Landscape of Extremes

Perhaps the most striking feature of the American power landscape is the sheer range in plant size. The median power plant in this country generates just 95 megawatts -- enough to power a small town, but a rounding error next to the coal and nuclear giants.

At the other end, the ten largest plants in the country -- a mix of hydro, nuclear, gas and coal -- each exceed 3,000 megawatts. The top ten alone account for roughly 39,000 megawatts, or about 3.6 percent of the national total. That is less capacity than the state of Florida, which has 58,000 megawatts.




The gap between the typical American power plant and the giants that actually keep the lights on continues to widen as small-scale solar and wind installations proliferate. Whether those thousands of small renewable plants can collectively replace the aging coal and gas behemoths remains the central question of the energy transition.